

PLLM: Private Multi-Party Inference and an Extensible Research Harness

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Why PLLM

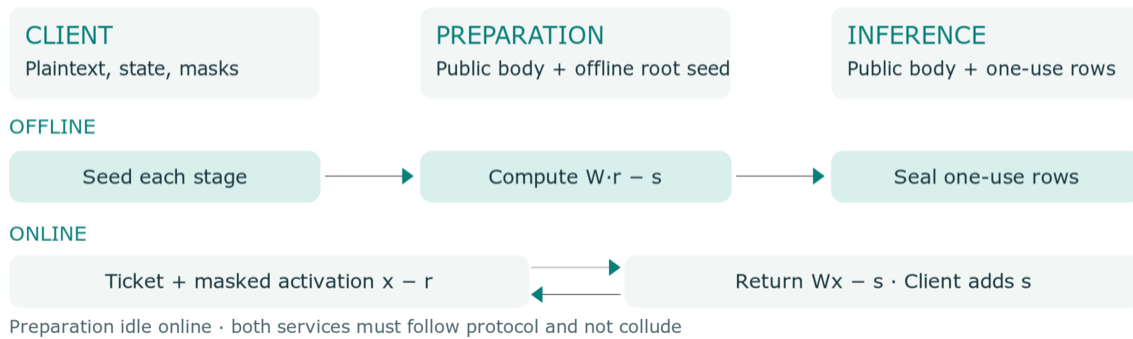
PLLM is a **high-performance private LLM multi-party inference runtime and extensible autonomous research harness**. A hosted endpoint ordinarily receives the request it computes on. PLLM instead aims to give separate operators useful work without granting either remote role the complete activation. Its second job: turn independently implemented research into compatible components that can be tested against a common baseline, rather than accepting paper-reported speedups across different models and trust assumptions.

High performance and economic value are **objectives**, not conclusions drawn from the present measurements. A faster isolated kernel is not automatically a faster, private, deployable model.

One private request

Client keeps prompts, tokenization, nonlinear and attention state, masks, sampling, and output. Before generation, trusted Preparation expands fresh stage seeds and uploads one-use masked corrections for public model matrices. Online, Inference receives a ticket and masked activation; Client reconstructs the result. Preparation is idle online. A trusted local gateway can expose Responses and Chat Completions APIs without turning the remote roles into application endpoints.

One response · distinct trust and execution roles



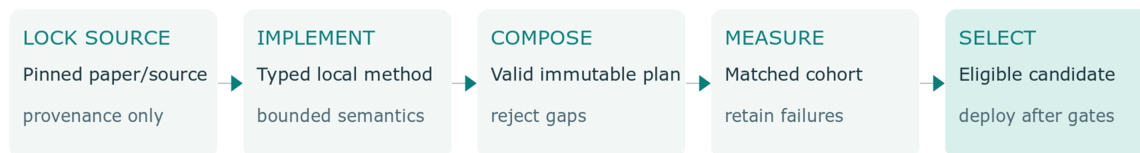
Offline and online work across Client, Preparation, and Inference. Each correction row is used once.

Preparation must follow the protocol, erase masks, and **not collude** with Inference; co-located loopback processes do not demonstrate independent operators. The current public path keeps attention, nonlinear work, and token boundaries local to Client. It does not establish malicious-provider security.

Research becomes a candidate capability

Researchers implement pinned papers through typed component or provider contracts. Immutable experiments bind model, composition, workload, and deployment. Bounded search proposes **valid** candidates; matched benchmarks retain failures. Functional, numeric, privacy, quality, and deployment checks gate adoption.

Research-to-capability loop · evidence grows, claims stay scoped



Checks: function · privacy contract · quality · trust · numeric policy
No automatic paper-to-code translation or measured SOTA winner yet

From paper provenance to a typed implementation, compatible plan, matched evidence, and an eligible deployment choice.

This is an autonomous *experiment-enabling* loop, not automatic paper-to-code translation or security certification. Papers are provenance, never executable dependencies. Static provider manifests support discovery; native provider runtime loading is not yet exposed. New research can improve a deployed choice only when an implementation passes these gates and beats its **eligible matched baseline**. No winner crosses different model, privacy, numeric, hardware, input/output-token, output-cap, or warm-state cohorts by assumption.

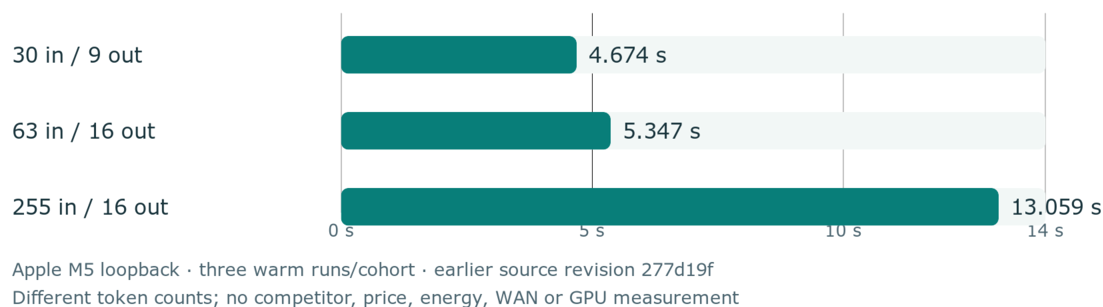
Economic value needs an honest denominator

A candidate can be economically useful if it meets the same privacy, model quality, and service constraints while reducing *total cost per successful response*: client work, offline preparation (including burned rows), online compute, network, and operator resources. Latency, throughput, traffic, and quality have different units; a single speed number cannot replace this accounting. PLLM represents metrics and matched benchmark results, but **has not measured prices, energy, or an economic return**, nor established an external state-of-the-art winner. Economic value is a testable hypothesis.

Measured baseline; open comparison

One retained study ran Client, Preparation, and Inference on an Apple M5 loopback host using a pinned Qwen2.5-0.5B-Instruct checkpoint. Three warm measurements per *different* input/output-token cohort give these full-response medians (including offline preparation). They are historical measurements from an earlier runtime revision, not current compiler or WAN results.

Historical full-response latency · Qwen2.5-0.5B



Historical Qwen2.5-0.5B full-response medians for three distinct token cohorts; no cross-system performance claim.

An optional verification component has only a separate, one-run **tiny-model** functional comparison. Garbling and cache transformations remain bounded or uncomposed, not whole-model Qwen speedups. Next: complete protected decoder coverage, deploy genuinely independent roles, and compare alternatives on the same checkpoint and workload with measured cost and quality. The technical paper records precise methods, paper-derived components, evidence, and missing comparisons.